Cheuk Hin Li, Robotics Engineer

Hong Kong, +852-65063916, marcoli434ovo@gmail.com

LINKS

Github, LinkedIn

PROFESSIONAL SUMMARY

Embedded / robotics software engineer focused on **STM32** firmware, **sensor data processing**, and **control**. Built the **climbing mechanism** and localization stack for HKUST's ABU Robocon robot; currently researching **magnetic tactile sensors** at Mevita Robotics Lab. **Seeking a 2026 winter internship in embedded or robotics software.**

EXPERIENCE

Jun 2026 — Aug 2026

Software Engineer Intern, Hong Kong Broadband

- Automated migration and validation of 20,000+ lines of financial data from on-premise ERP to NetSuite via **Python** and **CSV processing.**

Sept 2025 — Jun 2026

Embedded Software Developer, HKUST Robotics Team

- Develop embedded firmware in C for **STM32F407** microcontrollers, including **sensor interfaces** and **motor control**.
- Designed and implemented the **climbing mechanism** for War Dragon Robot.
  - Solved the **motor overheating** problem and **angle-rounding error** for climbing mechanism, increasing the climbing speed by over **121%**
- Collaborate with mechanical and hardware subteams using **Git-based workflows** for integration and code reviews.

TECHNICAL PROFICIENCIES

Languages:	C, C++, Java, Python, JavaScript
Embedded:	STM32F407, Arduino, CAN, I2C, SPI, UART, VESC
Robotics & Control:	PID, sensor fusion, state estimation, FSM
Tools:	Git, Bash, Linux, PyTorch (foundational)

EDUCATION

Sept 2024

BEng in Computer Science with Extended major in AI — Hong Kong

The Hong Kong University of Science and Technology

- Expected Graduation: Jun 2028
- Relevant Courses:
  - System Modeling, Analysis and Control
  - Reinforcement Learning
  - Honors Design and Analysis of Algorithms

PROJECTS

2025 — 2026

War Dragon

- **Competition Robot** for **ABU Robocon 2026** local competition
- Developed embedded firmware in C for STM32F407, handling sensor interfaces, motor command dispatch, and multi-axis coordination.
- Generated **trapezoidal velocity profiles** for the **rack-and-pinion** lifting mechanism, issuing **position setpoints** to VESC-driven motors.
- Leveraged **VESC**'s native position PID with FOC for **inner-loop control**, separating **trajectory generation** (STM32) from current/position loops (VESC).
- Diagnosed **CAN bus saturation** in a multi-motor setup; mitigated by **condensing feedback values**, sustaining 1 Mbps communication across over 15 nodes.
- Implemented **robot localization** fusing 6-axis IMU via **Madgwick filter** and positioning using magnetic encoders , achieving 0.4°-2° attitude drift per cycle and 0.4 - 1.1% position error per meter.